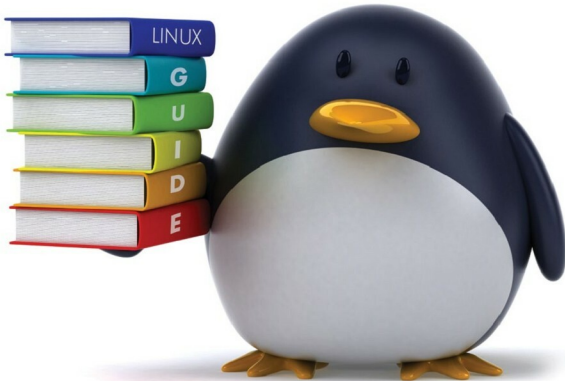


A Beginner's Guide to Linux

There are many valid and compelling reasons to opt for Linux as the operating system for a desktop computer, the most important being that it can be tailored to the user's needs and tastes. This article, aimed at noobs, describes the salient features of the Linux operating system.



Linux was designed for the sole purpose of providing desktop users with an affordable but functional operating system in a world where the most commonly used OS comes with a high price tag.

Linux was originally created in 1991 by Linus Torvalds, a Finnish student from the University of Helsinki, for his own personal computer, and with zero plans for expansion beyond one platform. However, Linux grew to become adaptable with several computer architectures and now runs on every platform imaginable.

Linux has fascinated developers ever since it was created and, as a result, its open source code has seen several contributions from more than 12,000 programmers.

The Linux kernel (the main part of the OS located at the core) has been released under the GNU General Public License (GPL) version 2.

Linux kernel architecture

The Linux kernel operates on a monolithic architecture, which basically means that the complete operating system works on the kernel space and on the solo supervisor mode. Monolithic

kernels have several pros and cons, some of which include:

Pros

- Direct and faster approach for hardware to access programs
- Simplicity and ease of in-process communication
- Quick processing of requests

Cons

- Lack of security due to processes running solely on supervision mode
 - Enormous memory and installation footprints
- However, kernel developers did a good job by overcoming the flaws associated with the monolithic kernel by modelling modules of the kernel, which can be both loaded and unloaded at any time. Modules that are able to support features such as low level virtualisation can also enable you to replace your kernel without rebooting the entire system, although this happens only in rare cases.

Common features of the Linux kernel

Let's look at some of the common features of the Linux kernel.

Multi-user support: A Linux system supports several